

130R-23: Demonstrating Entitlement to Cumulative Impact Claims in Construction

2 of 17

May 15, 2023

1. INTRODUCTION

This recommended practice (RP) of AACE International provides guidance on the process to demonstrate entitlement to cumulative impact claims on a construction project when a process for addressing cumulative impact claims is not addressed in the contract terms. This process applies across all common construction contracting strategies or delivery methods if the contract terms do not address any or some aspects of the procedure to address cumulative impact claims.

This RP is intended to provide guidelines (*i.e.*, not be a mandated standard) for a suggested process to use when submitting a contract change order request (COR) or claim associated with cumulative impact and what to look for when defending or analyzing such claims on construction projects.¹ This recommended practice represents the concepts most practitioners consider a good industry practice to use and recommend. This recommended practice is relevant to stakeholders on a construction project, whether owner, designer, contractor, subcontractor, construction manager, or others. Although this recommended practice is written in the context of a contract between an owner and prime contractor, it is applicable to any party contracted to perform work on a project, including contracts between prime contractors and their subcontractors and their suppliers.

The concept of cumulative impact has been recognized within the construction industry internationally for many years. It has also been referred to as ripple effect or knock-on effect and as a global claim in English literature and case law. Cumulative impact is defined as the net impact of two or more undifferentiated changes, as each is measured or measurable at a certain point in time, being much greater than the sum of the effect of the individual parts. This effect results in the reduced productivity of unchanged work. Due to the complicated nature of construction work, it is not usually well understood by contracting parties and even legal professionals. Cumulative impact, when it occurs on a construction project, is often referred to as an indirect disruption or loss of productivity claim. There is no specific number or total value of changes on a project that, once reached, create the circumstance for a cumulative impact. The terms of most construction contracts allow for changes, but they do not typically define any maximum limit for changes. This lack of a contractual limit can lead to disputes regarding whether the number or value of changes was reasonable or foreseeable.

This RP provides a basic understanding of cumulative impact and outlines the steps necessary to demonstrate or refute a contractor's entitlement to the potential damages that may result. Cumulative impact is defined as:²

- The unforeseeable disruption of productivity resulting from the *synergistic* effect of an undifferentiated group of changes. Cumulative impact is referred to as the *ripple effect* of changes on unchanged work that causes a decrease in productivity and is not analyzed in terms of spatial or temporal relationships;³ and
- The impact on unchanged work, which is not attributable to any one change but flows from the synergy of the number and scope of changes issued on a project.⁴

In a cumulative impact claim, the aggrieved contractor seeks compensation for the collective damage caused by all of the changes in addition to particularizing the damages arising from each change. Cumulative impact claims are often erroneously called total cost claims. *Total cost* describes a method for quantifying cumulative impact claims not the actual condition per se.

May 15, 2023

1.1 Productivity Loss and Disruption

To understand disruption, it is necessary to understand what it impacts. In the context of a construction project or in manufacturing, disruption impacts productivity. Productivity is defined as:

A measure of output relative to input. Productivity (or efficiency) is improved by increasing output for a given input or decreasing input for a given output. If the input is specifically work hours, the term commonly used is labor productivity.⁵

Productivity loss can occur when a contractor does not accomplish its planned production rate.⁶ In the context of trade labor hours and equipment hours, productivity loss can be described as a loss of efficiency with respect to contractor producing less than its planned output per work hour of input. Thus, the contractor is expending more effort (hours) per unit of production than originally planned, assuming its planned productivity was reasonably estimated.⁷ Expending more hours than was planned typically means expending more money to complete the work. Loss of productivity is caused by a detrimental change in or 'disruption' to planned resource usage, working conditions, or work method, which typically results in increased costs.⁸ Productivity losses against the planned rate may also occur if the contractor performing the work is overly aggressive on its estimate of planned performance or various other reasons that are within the control of the contractor.

Productivity involves both labor and equipment efficiency. Some types of disruption claims pertain to equipment disruption, such as in earthwork, mining, and road construction projects. The damages sought for equipment productivity loss may be in the form of additional equipment operators and equipment required or for the cost of the extended duration of the equipment operators and equipment usage, including added standby time.

Disruption is defined as:

An interference (action or event) with the orderly progress of a project or activity(ies). Disruption has been described as the effect of change on unchanged work and manifests itself primarily as adverse labor productivity impacts. If such disruption is caused by owner or engineer action (or failure to act), the contractor may be entitled to recover any resulting costs.⁹

Therefore, disruption is caused by a change to some project dynamic that a contractor ultimately experiences as a negative impact on the planned productivity for work that had not been previously changed. Disruption can impact a contractor's costs and/or schedule.

Parties to a construction project often focus on specific construction site aspects and may lose sight of the interfaces and dynamics between project phases and locations of work. In order to address the full scope of work one must take into account disruption to:

- Engineering
- Procurement (suppliers)
- Fabrication (may be at multiple locations globally)
- Logistics (trains, planes, trucking, sea cargos, etc. - again, often globally)
- Direct work at construction site(s)
- Subcontracted work at construction site(s)
- Indirect labor at field office (site) and home office

May 15, 2023

1.2 Direct and Indirect Disruption

Disruption can be further separated into two primary categories: either *direct* or *indirect*.

Cumulative impact is described as an indirect disruption. Before indirect disruption is described, it is necessary to explain direct disruption. Direct disruption, also known as local disruption, is defined as:¹⁰

- The direct impact that changed work has on other unchanged work going on around it;¹¹ and
- The immediate and direct impact resulting from change or other circumstances that lowers productivity in the performance of the changed or unchanged work. Direct impact is considered foreseeable and the disrupting relationship to unchanged work can be related in time and space to a specific change.¹²

Direct disruption affects unchanged work that was temporally and physically near the disrupted work. That is, direct disruption typically affects work occurring simultaneously with or soon after the disruption event. The associated costs to the contractor that result from direct disruption relate to the direct and foreseeable consequences of a change.¹³ Refer to AACE Recommended Practice 25R-03 for a list of common causes of lost productivity.¹⁴

Contrary to direct disruption, indirect disruption, or cumulative impact, does not necessarily affect unchanged work that is temporally or physically close. Also, direct and indirect disruption do not always result from many changes. It is up to the analyst to identify and explain the disruptions and relate to what occurred, including an assessment of the timing of the changes. For example, if most of the changes occurred during the latter part of a project, reduced productivity during the first part of the project most likely does not result from the ripple effect of the later occurring changes.

A common cause asserted for cumulative impact is that it was the result of multiple changes whose impact could not be determined until much later in the project or after the project had been completed. Cumulative impact has metaphorically been described as the disruption caused by a 'thousand cuts'. An individual change does not cause indirect disruption; rather, indirect disruption may be caused by the effect of *multiple* changes. Multiple requests for information (RFIs) can cause cumulative impact in certain circumstances, but an extraordinary number of RFIs alone cannot demonstrate the contractor's entitlement to recovery of its additional direct labor and equipment costs based on cumulative impact.

One way to understand the occurrence of cumulative impact on a project is illustrated by the following decisions:

*This phenomenon arises at the point the ripples caused by an indivisible body on two or more changes on the pond of a construction project sufficiently overlap and disturb the surface such that entitlement to recover additional costs resulting from the turbulence spontaneously erupts.*¹⁵

*...[t]he underlying theory is that numerous changes cause a cascading ripple-type of impact on performance time and efficiency which is too uncertain or diffuse to be readily discernable at the time of pricing each individual change.*¹⁶

¹⁰ See AACE International Recommended Practice 10S-90, Cost Engineering Terminology.

¹¹ Centex Bateson Constr. Co., VABCA Nos. 4613, 5162, 5165, 99-1 BCA ¶ 30,153 (1998), *aff'd*, Centex Bateson Constr. Co. v. West, 250 F.3d 761 (Fed Cir. 2000) at 149,258 [citing Triple "A" South, 94-3, BCA ¶ 27,194, ASBCA No. 46,866 (1994) at 135,523].

¹² Change Orders Productivity Overtime – A Primer for the Construction Industry, MCAA, 2012, p. 82, citing Centex Bateson Constr. Co., VABCA No. 4613, 5162, 5165, 99-1 BCA ¶ 30,153.

¹³ Long PE, Richard J. et al, Cumulative Impact and Other Disruption Claims in Construction, 2014, Virtualbookworm.com Publishing, Inc., College Station, TX., §1.1.2, p. 6.

¹⁴ See AACE International, Recommended Practice 25R-03, Estimating Lost Labor Productivity in Construction Claims, pp. 4-7 of 29.

¹⁵ Centex Bateson Constr. Co., VABCA Nos. 4613, 5162, 5165, 99-1 BCA ¶ 30,153 (1998), *aff'd*, Centex Bateson Constr. Co. v. West, 250 F.3d 761 (Fed Cir. 2000) at 149,258.

¹⁶ McMillin Bros. Constructors, Inc., EBCA No. 328-10-84, 91-1 BCA ¶ 23,351.

May 15, 2023

An impact claim—often characterized using other names, such as, ‘cumulative impact,’ ‘ripple effect,’ ‘loss of labor efficiency,’ or ‘loss of productivity’—is based upon the theory that individual compensable changes to a Contract, taken as a whole, can have such a disruptive effect on the contractor’s performance that the contractor has a compensable claim for costs in addition to the amounts of its individual change orders.¹⁷

Cumulative impact need not be traced to specific causes of increased performance costs, but can arise from changes which, when viewed retrospectively, were so many and had such effect on performance that there is a separately compensable impact claim. We do not require the specificity here that would otherwise be necessary for Appellant to prevail on entitlement. The sheer number of change orders, however, cannot establish the validity of a cumulative impact claim. Impact is not demonstrated solely by showing the number of changes or clarifications to the contract. General unsupported statements that a contractor suffered impact are not sufficient proof that it did, and the absence of contemporaneous documentary evidence of the disruptive effects to the work may provide reason for rejecting the claim. ... There must be evidence of the number, timing, and effect of the changes that were issued.¹⁸

In summary, because cumulative impact is difficult to identify and quantify contemporaneously due to the nature of impacts caused by multiple change events, it is challenging to precisely pinpoint causation and the resulting damages. However, cost increases and delays resulting from changes to previously unchanged work and disruption are very real and often very significant.

2. RECOMMENDED PRACTICE

The purpose of this recommended practice is to suggest an analysis process by which a claimant (typically the contractor) can include within a contract change order request (COR) or claim that requests damages in the form of additional cost and/or time resulting from cumulative impact. This RP also addresses areas that an owner should be aware of with respect to evaluating such claims on construction projects.

Cumulative impact typically manifests well after the disruption has occurred and contractors are typically not adequately prepared to demonstrate how or when the disruption occurred. Before the project starts it is important that the contractor establishes adequate project control processes to identify as quickly as possible potential negative variations in productivity that maybe the result of inexplicable causes. When it occurs, most contractors recognize that their labor and equipment productivity is decreasing, or their labor and equipment unit costs are increasing. Unfortunately, some contractors fail to maintain adequate contemporaneous jobsite records to allow for evaluation of impacts to labor and equipment productivity and the costs of other impacts. Since cumulative impact typically cannot be identified contemporaneously, contractors often do not realize that they are incurring impact costs until significant cost increases are identified near or after project completion. Because claimants normally bear the burden of demonstrating liability, causation, and quantum,¹⁹ the failure to maintain adequate records and contemporaneously identify impacts can be a barrier to recovery of costs associated with cumulative impacts. Still, recovery may be possible if efforts to manage changes were reasonable and the full cumulative impact of changes could not have been determined contemporaneously. Reservation of rights in the underlying change orders may also be an important factor affecting the ability to recover cumulative impact costs beyond those in the contemporaneous change orders.

It is recommended that a change management process be agreed upon before the work commences because experience has shown that one cannot go into projects assuming a perfect world whereby:

- The owner never changes its mind.
- The architect/engineer never alters or errs with its design documents.

¹⁷ Jackson Const. Co., Inc. v. United States, 62 Fed. Cl. 84, 103 (2004).

¹⁸ Appeal of Bechtel Nat., Inc., NASABCA No. 1186–7, 90–1 BCA ¶ 22,549 at 113,177.

¹⁹ See Section 2.2 for further discussion regarding demonstrating entitlement.

May 15, 2023

- The contractor never performs poorly or mismanages its work.
- Forces outside the control of the contracting parties are foreseeable and manageable.

That perfect world does not exist in the context of capital construction projects.²⁰

Although claims for cumulative impact have been submitted on projects located in the U.S. and other countries, demonstrating entitlement to this type of impact has proven to be quite difficult. There are no known detailed processes or methodologies that can be replicated to reliably demonstrate entitlement, primarily because the circumstances that lead to cumulative impact are complex and unique to each project and require project-specific analysis.

2.1 Demonstrating Entitlement to a Change Order or Claim

Before the steps associated with demonstrating entitlement to a cumulative impact change order or claim can be discussed, it is necessary to first describe the over-arching requirements to demonstrate entitlement to a contract change. A contract change is defined as an:

*Alteration or variation to the project scope of work, price, and/or the schedule for completing the work.*²¹

In the engineering and construction industries, the primary change management tool used to document and authorize changes to the contract is the *change order*, also referred to as a *variation order*.

A contract change order is defined as:

*A document requesting and/or authorizing a scope and/or baseline change or correction. 1) From the owner's perspective, it is an agreement between the project team and higher authority approving a change in the project control baseline. 2) From a contractor's perspective, it is an agreement between the owner and the contractor to compensate for a change in scope or other conditions of a contract. It must be approved by both the client and the contractor before it becomes a legal change to the contract.*²²

A change order may address only one change or can be a consolidation of several seemingly independent changes.

If a dispute arises regarding the quantification of damages resulting from cumulative impact that is identified in a change order request during the project (or after cumulative impact was identified after the project had concluded), then the contractor (claimant) may resubmit the issue as a claim later in the project or, more commonly, after the project has completed. The contractor may need to further supplement the change order with additional information to support its cumulative impact claim. This may be reasonable if the full extent of the damages requires a complete assessment of the increased labor and equipment hours through the entire impact period. It is an entirely separate issue to raise the cumulative impact claim after the project is completed, which could be rejected due to late notice (see section 2.2.2 below), unless the contractor had expressly reserved rights for the cumulative impact. Therefore, the contractor should attempt to identify these types of issues and quantify them during the project so that they can better pursue recovery and eliminate notice objections by the owner. Doing this work after the project is complete should be a last resort.

A claim is defined as:

May 15, 2023

A demand or assertion of rights by one party against another for damages sustained under the terms of a legally binding contract. Damages might include money, time, or other compensation to make the claimant whole.²³

Demonstrating entitlement for a disputed claim can be a rigorous, expensive, and time-consuming endeavor. Therefore, the claimant should carefully consider whether pursuing a claim makes the best business sense before initiating this process or how far to pursue it within the dispute resolution process.

2.2 Elements for Demonstrating Entitlement in a Cumulative Impact Claim

Cumulative impact claims are among the most difficult types of claims to demonstrate entitlement for recovery of damages. The reason is that loss of productivity associated with cumulative impacts cannot be easily observed, particularly when the impacts are to work that has not yet started. Nor is it easy to apportion responsibility for the quantum of time and cost impacts when multiple issues have a role in causation.

Recommended Practice 120R-21²⁴ identifies seven primary elements a contractor should address to demonstrate its entitlement to a change order request or claim. These seven primary elements include:²⁵

1. Causal event occurrence.
2. Adherence to contract change notice requirements.
3. Contractual entitlement to request a contract change or make a claim.
4. Causation.
5. Prudent effort to mitigate any impacts.
6. Assignment of responsibility.
7. Quantification of impacts.²⁶

These elements describe the basic considerations that should be addressed to demonstrate entitlement to a change order or claim, including claims for cumulative impact. Addressing these specific elements of entitlement is recommended because one or more of these elements are often the focus of disputes if not adequately addressed by the claimant in its change order submittal. In addition, applicable contract law in the contractual legal jurisdiction should be evaluated for the necessary entitlement requirements. Another consideration is the type of contract that is agreed upon by the parties, as reimbursable contracts may be treated differently than lump-sum or fixed-price contracts, depending on other terms in either type of contract. Therefore, it is recommended that parties to a claim or contract dispute consult their attorneys to consider their legal options.

In the following section, the seven elements of entitlement are discussed in the context of a cumulative impact claim:

2.2.1 Causal Event Occurrence

The causal event in the context of this recommended practice is demonstrating the occurrence of cumulative impact. Before determining whether there was a resulting impact (cumulative impact) on the project, the claimant must first demonstrate that the conditions existed on the project, which led to the resulting impact. The ultimate cause may be changed work, which may be difficult to document through contemporaneous records. Demonstrating that indirect disruption has occurred is one of the most difficult elements to demonstrate because it is not foreseeable or obvious until much later in the project after the future unrelated work was found to have been impacted.

130R-23: Demonstrating Entitlement to Cumulative Impact Claims in Construction

8 of 17

May 15, 2023

Cumulative impact does not occur as the result of a single event or occurrence. Rather, cumulative impact results from *"the synergistic effect of an undifferentiated group of changes."*²⁷ It is often not until a cause-and-effect relationship (causation, the demonstrated reason(s) causally connected to the increased labor and equipment hours) has been identified that cumulative impact can be demonstrated to have occurred. Proving that cumulative impact occurred requires the ability to identify the symptoms or effects of the impacts, often before recognizing a cumulative impact occurred. Therefore, it is important, and to the contractor's benefit, to plan for and track labor and equipment hours at the most detailed level possible and incorporate the individual changes into the project schedule.

Demonstrating that the causal events took place can be shown by plotting productivity over time. One approach is to use labor hours plotted against performance in the form of percent complete, which can be pulled out of a schedule with the right level of detail. The expenditure of hours over time, shown as a percentage complete against the budget with proper adjustments resulting from executed changes, will show when productivity trends change and is useful in identifying the beginning of disruption impacts on productivity, the steady-state of productivity, and the easing of impacts on productivity. This is also a way to identify and quantify (or exclude from the quantification claim) the typically inefficient ramp-up and ramp-down areas of the project that are associated with inefficient work resulting from learning curves and completion of the punchlist (punchout).

2.2.2 Adherence to Contract Change Notice Requirements

One unique aspect of cumulative impact is that contractors who have experienced this phenomenon typically do not realize it has impacted their work until well after the causal events have occurred. This circumstance can create a conundrum for the contractor if it did not preserve its rights to submit a change notice, change order request, or claim months or even years after the causal events occurred.

In many construction contracts between owners and contractors, the change management clause requires that the contractor submit notices and change requests as well as the required change documentation within a specified number of days after the contractor recognizes a change or after the change event has occurred. Because contractors may not recognize the cumulative impact until months or years after it occurred, owners may assert that a cumulative impact change request or claim was submitted late - therefore exceeding the specific contract notice requirements - and as a result, reject the change request or claim.

However, if and when a contractor recognizes that its productivity has been affected by cumulative impact, it should submit a timely written notification to the owner.

*Once the contractor becomes aware that a multitude of changes are impacting its productivity, i.e., the impacts are foreseeable, it is essential that the contractor notify the owner of its findings, and that the contractor reserve its rights to claim such impacts on any subsequent change orders at the end of the project.*²⁸

Depending on the specifics of a case, one may find a contractor not entitled to damages arising from the cumulative impact absent proper notices. If allowed by the parties' contract, a written reservation of rights by the contractor in a change order request (which is also incorporated into the approved change order) can be an important tool for the contractor if done correctly and may lead to its contractual entitlement to additional costs.

One court has stated, however, *"Although there is no need to engage in a pre-Contract Disputes Act analysis of*

May 15, 2023

whether cumulative impact is a separate constructive change or a cardinal change in order to preserve our jurisdiction, it seems to us that cumulative impact must be viewed as a separate constructive change compensable under the CHANGES clause. This constructive change, although resulting from them, is independent of other contract changes and can survive an accord and satisfaction on a change, or a general release.”²³

This recommended practice is based on the premise that cumulative impact is considered a constructive change.

The subject of reserving rights to claim for cumulative impact is a legal one. Therefore, it is recommended that all parties consult with their attorneys before executing a contract to ensure all rights are appropriately addressed and a common agreement is reached before executing a contract.

2.2.3 Demonstration of Contractual Entitlement

The contractor must demonstrate that it has a contractual right to assert the cumulative impact claim and that it was not the result of impacts that were its responsibility. That is, the contractor must demonstrate that the changes were caused by the owner or others; i.e., they are constructive changes. The contractor has a burden to substantiate its case and the obligation to set forward sufficient evidence to be considered by the owner or trier of fact.

2.2.4 Demonstration of Causation

The contractor must demonstrate the cause-and-effect relationship between an impact that was not its responsibility and the effects that the impact had on its productivity, costs, and ability to complete the project according to schedule. Correlation between impacts and effects alone does not suffice to demonstrate causation and claims that simply show correlation do not tend to succeed.

The causation element is often the most difficult to demonstrate entitlement, especially as it concerns demonstrating a cause-and-effect relationship between the changes or events that caused the cumulative impact. Even if multiple changes are suspected to be the cause of cumulative impact, that, in and of itself, does not mean cumulative impact exists. If the contractor suspects that there was cumulative impact on its project, it often must drill down further to demonstrate that the multiple changes contributed to the effects that resulted in the indirect disruption.

The focus should be placed on well-defined groups of change orders that caused the indirect disruption rather than a handful of individual change orders. Often, for example, the cause of the disruption can be found in changes associated with a particular discipline on a project or changes that occurred over a particular period of time. Focusing on changes that may have affected one discipline or time period is recommended. However, it is not uncommon for changes to one discipline to have a ripple effect on the follow-on work of another discipline. For example, direct changes that impact plumbing could have a subsequent impact on electrical or sheetrock trades. Other issues regarding scope changes that could result in indirect disruption include:

- Shared resources, including labor and equipment.
- Shared work areas; stacking trades.
- Timing of changes.
- Scope reduction.
- Scope increases.
- Changes to owner-furnished equipment.
- A generally excessive number of change orders.

130R-23: Demonstrating Entitlement to Cumulative Impact Claims in Construction

10 of 17

May 15, 2023

- Lack of scope definition.

Any or all these types of circumstances and events can contribute to cumulative impact. Demonstrating a causal (cause and effect) relationship between multiple changes that caused the contractor's increased labor and equipment hours and costs (the contractor's recoverable damages) and the total increased labor and equipment hours and costs is typically the most difficult task for a contractor to prove entitlement, especially with a cumulative impact claim. Damages may be in the form of cost and/or schedule impacts. Although multiple changes may have been executed on a project, this fact is typically not enough to demonstrate that cumulative impact occurred.

The added difficulties for a contractor to demonstrate causation on a cumulative impact claim are the impacts to work scope that may not have been temporally or geographically near the time and location of the impacting changes.

To demonstrate causation, the contractor must first establish the number of change events was so excessive that it resulted in cumulative impact. As noted previously, the quantity of changes alone does not establish a claim's validity. Rather, a combination of the quantity and the total cost (specifically labor and equipment hours) of the change orders compared to the base contract may be more indicative that a cumulative impact has occurred.³⁰ The impacts on productivity must be severe enough that it is clear the changes were responsible for the indirect disruption.

In addition to the consideration of multiple change events contributing to the cause, there are myriad factors, some more complex than others, that may affect how or if labor and equipment productivity were affected as a result of cumulative impact. Factors that may affect the difficulty of demonstrating a cumulative impact claim include project complexity, the relationship between productivity and the project schedule, and the timing of the impacting changes. Therefore, the contractor should explain these factors in the context of how they influenced productivity disruption. In addition to analyzing the multiple change orders that were issued and the labor and equipment data, the contractor must use other contemporaneous documentation to demonstrate causation, such as schedules, monthly and weekly progress reports, and other documentation to describe what was going on in the field while the deficiencies in labor and equipment productivity were taking place and why they were taking place.

It is important to understand that there is no simple formula to demonstrate causation. Courts and boards have acknowledged the difficulty in demonstrating the (causal) link between changes and loss of productivity.³¹

The contractor must demonstrate in its causation analysis that neither they nor any of their subcontractors or suppliers were responsible for the productivity loss and that no concurrent impacts for which it was responsible occurred. If they were, the contractor's claim calculations should have allocated the increased labor and equipment hours and costs for the contractor's contribution to the cumulative impact. The contractor must also demonstrate to an acceptable level of certainty that it did not contribute to the loss in productivity relative to the original scope and approved change orders by, amongst other things:

- Underestimating its labor and equipment hours.
- Underestimating its productivity rates.
- Mis-scheduling the works.
- Mismanaging the affected works.

Simply showing a large number of changes and a corresponding cost overrun is typically not enough. Some meaningful connection or correlation between changes and cost growth must be established. Explaining how changing a light switch's location in the east basement can impact work in other parts of the building is not an easy

May 15, 2023

task. Likewise, explaining how adding an additional pipe in one process unit can impact the electrical work in another unit is not easy, and one must remember that the contractor bears the burden of proof.³²

Figure 1 below is a simple cause and effect diagram that illustrates the process flow from the potential influences that lead to indirect disruption to what the contractor might experience ultimately as cumulative impact. This graphic is a simple and easy way to communicate how the cause-and-effect nature of indirect disruption in the field can lead to cumulative impact. The black dots in Figure 1 represent the cause-and-effect relationship between items in one leg to items in the next leg.

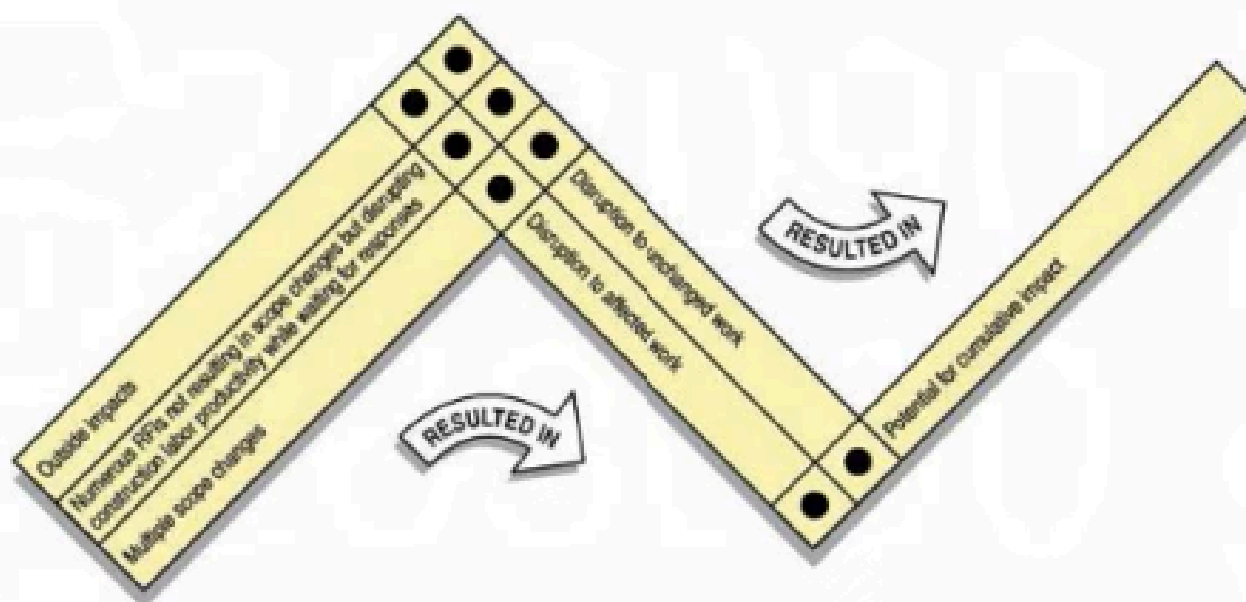


Figure 1 - Simple Cumulative Impact Cause and Effect Diagram

2.2.5 Prudent Effort by the Contractor to Mitigate Any Impacts

Because the occurrence of cumulative impact is unforeseeable, there is typically little for a contractor to do proactively to mitigate any indirect impacts. However, the primary focus for a claimant is to keep accurate and complete labor, equipment, productivity records, as well as contemporaneous daily progress narratives that describe the reasons for the disruption or deficiency. By doing so, the contractor should be able to contemporaneously and quickly identify the occurrence of any inexplicable productivity impacts to work that may have been caused by a cumulative impact, and then promptly provide written notice to the owner pursuant to typical contractual language.

2.2.6 Assignment of Responsibility

Identifying responsibility for the party responsible for a cumulative impact claim can be difficult. Because it is difficult to demonstrate causation, even if the approved change orders that led to a cumulative impact are clearly known, it can be difficult to demonstrate that the contractor was not responsible for its own loss of productivity. It may be necessary to allocate responsibility for cumulative impact productivity loss between the owner and contractor if the contractor also caused impacts that negatively affected the productivity of its work.

May 15, 2023

If the contractor identifies that it was partially culpable for any of the loss in productivity associated with a cumulative impact claim, it would be prudent to acknowledge this responsibility in its claim. It could be a significant disadvantage to the contractor if the owner finds productivity loss for which the contractor was responsible, especially if in preparation for a trial or arbitration, as this could lead to the adjudicators calling into question the contractor's motives for submitting the claim and the validity of the contractor's productivity loss calculations.

2.2.7 Quantification of Impacts

The contractor must demonstrate that the cost or schedule impacts that have been or will be realized due to the occurrence of cumulative impact are reasonable, accurate, and supported.

It must be understood that loss of productivity may not be directly observed in the field like other events that would create the need for a contract change. Productivity loss can only be identified and quantified by analyzing labor and equipment hours and related cost data and considering the project dynamics affecting the project before, during, and after the impacted work has been performed.

Because cumulative impact is most often an impact on labor productivity, the claimant must estimate and quantify lost productivity through analysis. Then, it must connect that loss of productivity to the additional costs incurred (damages) that resulted.

Multiple changes that have a cumulative impact on work are often labor-related changes. Therefore, in assessing cumulative impacts, the dollar value of the changes that have occurred may not accurately describe their impact. Rather, the number of labor hours needed to perform the change is of particular importance in evaluating cumulative impacts because a main objective of a cumulative impact claim is to demonstrate the extent of loss of labor productivity arising from the synergistic effects of multiple changes.

Demonstrating entitlement to a cumulative impact claim is data intensive and requires significant analyses. The contractor must demonstrate through analysis that its work was either delayed and/or performance was made less efficient because of multiple changes or other influences that occurred on the project. A credible schedule delay analysis will need to be performed if the contractor was also delayed by the cumulative impact or other disruptive events. Relying on a productivity loss methodology alone will not be adequate for the recovery of delay damages.

There are several recognized methodologies that are intended to estimate and quantify the loss of productivity on a construction project. Some methodologies are preferred more than others. The methodology selection is often a consequence of labor hour and productivity data availability. There are several methodologies that have been recognized within the construction industry. Within Recommended Practice 25R-03, *Estimating Lost Labor Productivity in Construction Claims*, many of these methodologies (but not all) for estimating lost productivity are noted in the following:³³

These methods are listed in order of preference. The recommended order of preference of the applicability of the studies and methods set forth below is based upon the weight of published literature. That is, Project Specific Studies are preferred to Project Comparison Studies. Project Comparison Studies are likely to be given greater weight than Specialty Industry Studies. Specialty Industry Studies are generally considered more reliable than General Industry Studies, and so on and so forth. Within each category, this Recommended Practice has likewise placed the methodology in order of preference. For example, properly performed measured mile studies are preferred to earned

May 15, 2023

value analyses which, in turn, are considered more credible than work sampling or craftsmen questionnaires. ...

- **Project Specific Studies**
 - Measured Mile Study
 - Earned Value Analysis
 - Work Sampling Method
 - Craftsmen Questionnaire Sampling Method³⁴
- **Project Comparison Studies**
 - Comparable Work Study
 - Comparable Project Study
- **Specialty Industry Studies**
 - Acceleration
 - Changes, Cumulative Impact and Rework
 - Learning Curve
 - Overtime and Shift Work
 - Project Characteristics
 - Project Management
 - Weather
- **General Industry Studies**
 - U.S. Army Corps of Engineers Modification Impact Evaluation Guide
 - Mechanical Contractors Association of America
 - National Electrical Contractors Association
 - Estimating Guides
- **Cost Basis**
 - Total Unit Cost Method
 - Modified Total Labor Cost Method
 - Total Labor Cost Method
- **Productivity Impact on Schedule**
 - Schedule Impact Analysis

Figure 2 below is a graph that indicates several of the more familiar productivity loss analysis methodologies and evaluates and compares how the construction industry typically views the reliability of each. The factors that affect reliability include:

- a) Availability and quality of the contemporaneous documentation necessary to perform each analysis,
- b) The effort necessary to perform the analysis,
- c) The expertise and cost to prepare or use the analysis, and
- d) The risk (uncertainty) associated with convincing adjudicators of the reasonableness and reliability of the results.

Moving from left to right on the graph, more effort is necessary to perform each analysis methodology and uncertainty in the results decreases. That is, the cost, effort, and expertise required to prepare a reliable loss of productivity analysis typically increases as the methodologies are listed from left to right on the graph.

May 15, 2023

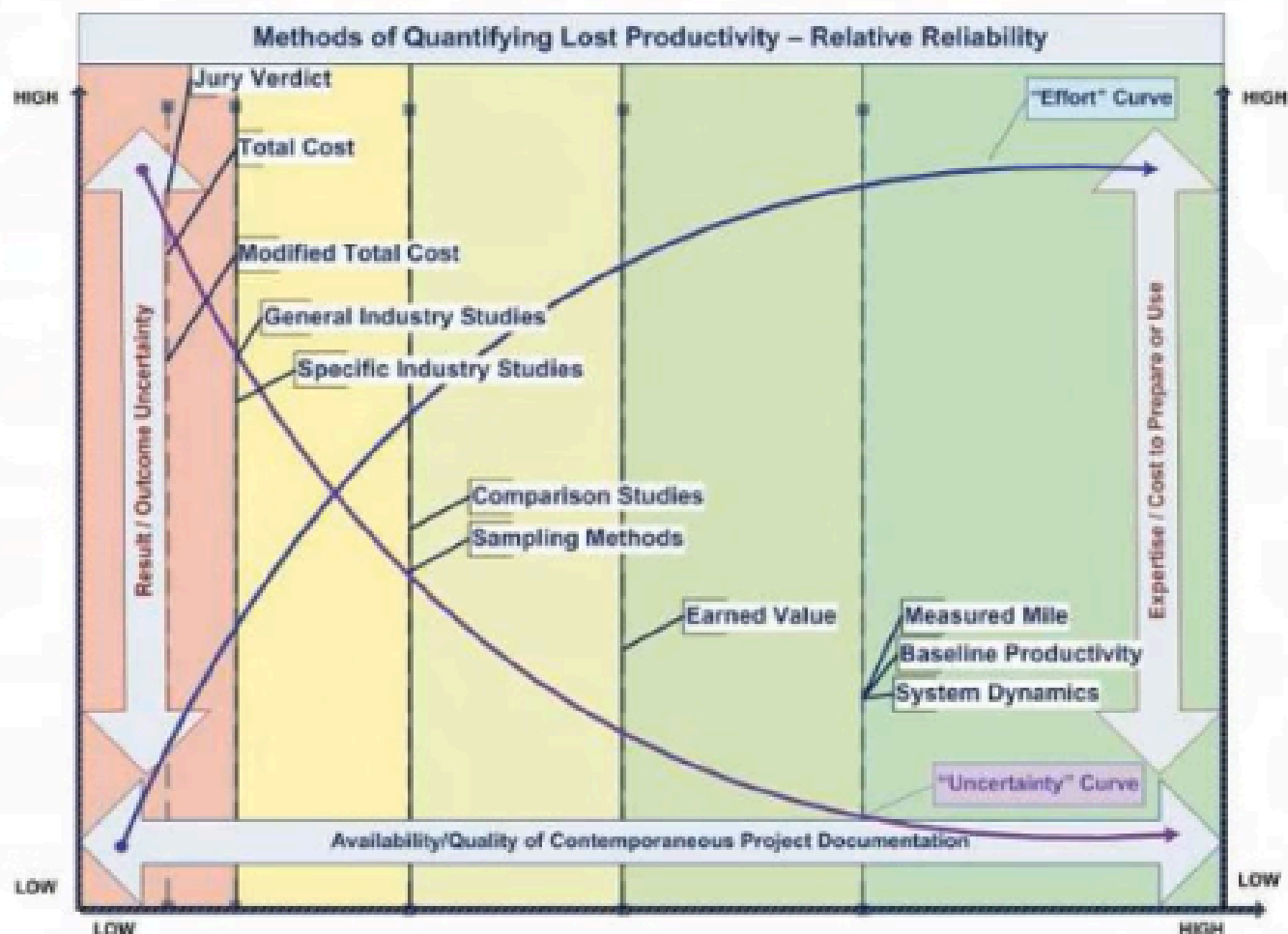


Figure 2 – Methods of Quantifying Lost Productivity – Relative Reliability²³

As the methodologies to calculate productivity loss are considered from left to right in Figure 2, it becomes increasingly important for the contractor to keep and maintain adequate man-hour and productivity data throughout the project.

The terms of the contract will play a key role in how entitlement is dealt with when seeking to apportion cost or time damages.

2.3 Supporting a Cumulative Impact Claim – Document Retention and Record-Keeping

The challenge for a contractor is clearly showing how its productivity was impacted by myriad changes using contemporaneous project records and data. When setting up the document control effort, it is helpful to consider all the information that will be vital to proving a cumulative impact claim. Specifically, the data collected should include location coding for each trade work so that it is easy to separate a specific trade work in one area from another. Then when impacts are shown to affect one area, the productivity can be easily compared to another area with no or fewer impacts. A master area designation plan with locations coded to the plan can be very helpful and focus data collection on locations.

130R-23: Demonstrating Entitlement to Cumulative Impact Claims in Construction

15 of 17

May 15, 2023

It may be difficult to demonstrate to a reasonable degree of certainty a claim for the cumulative impact of change on unchanged work if the contemporaneous record does not demonstrate the occurrence of disruption to unchanged work through the record-keeping discussed above. Again, this is not only a productivity quantification exercise, but one of contemporaneously documenting that unchanged work experienced additional mobilizations, was performed out-of-sequence, experienced idle time, increased work difficulty, etc.

However, claim preparation should not be left for the project close-out phase or other post-construction phases because the chances of collecting supporting contemporaneous evidence, retrieving backup documents, and verifying commitments from those individuals who have first-hand knowledge of the matters in a dispute are typically higher during construction once the claim is fresh. In post-construction, commitments are often forgotten, and locating backup documents or evidence may be challenging because project records may not have been organized properly, or project personnel may have been released or might be hard to reach out to.

If a dispute evolves into being adjudicated within an arbitration or trial setting, written factual evidence and analysis based on contemporaneous data should be relied upon more heavily than other forms of evidence, including testimony from fact and expert witnesses. Possessing accurate and well-maintained labor, equipment, and productivity records will significantly improve the contractor's chances for recovery. Also, ensuring that worker hour data is collected at an adequate level of detail, preferably at the lower levels of the work breakdown structure (WBS), will help align the claim with cost categories and a cost breakdown structure. WBS alignment helps map worker data with schedule data to improve the ability to perform the productivity loss analysis methodologies that are most often preferred. This mapping is hard to perform later without the correct detailed data, so the system should be set up to record labor data so that production records and schedule records align. A good organization that links labor costs, production quantities, schedule timing, and locations will ensure the ability to develop a convincing cumulative impact claim.

To enhance the likelihood that disputes can be avoided, it is important that the contractor, subcontractors, and owner maintain a document control system contemporaneous with project execution. Populating the document control system should begin before the notice to proceed and contain procurement documents (including bid documents and productivity assumptions), the contract, the baseline schedule, and productivity assumptions.

Adequate documentation is vitally important to the contractor interested in survival. First, it provides the information needed to effectively bid, plan, and construct the work. Second, it provides the hard data needed to analyze and successfully resolve the construction claim. A presumption of reasonableness arises in favor of such records that satisfies the claimant's burden and shifts it to the other party, who must then prove that the records are unreasonable and/or inaccurate.³⁶

Most large engineering and construction companies use document control systems to organize and maintain project records. However, experience has shown that the smaller contractors that experience cumulative impact are less likely to possess a robust document control system. Therefore, it is beneficial for prime contractors to encourage and assist their subcontractors to establish and maintain an adequate document control system. Owners and their construction managers should maintain a document control system throughout a project.

If a document control system is shared but managed by the contractor, the owner should consider downloading all documents during the project to its own document storage system so that in the event of a major dispute, it will still have access to the documents that it needs to evaluate the contractor's claim at the end of the project or during subsequent arbitration or litigation.

Labor, equipment, and productivity records should be kept, as well as other important documentation, including schedule updates, correspondence, progress reports, meeting minutes, change order requests, work package

May 15, 2023

planning, and any other documentation that will help support the seven elements necessary to demonstrate entitlement. Detailed project cost reports, productivity reports, or an earned value management system can be used to monitor the project and demonstrate that there were variances or overruns that were not accounted for in previously identified specific direct changes.

REFERENCES

1. AACE International, Recommended Practice 105-90, *Cost Engineering Terminology*, AACE International, Morgantown, WV (latest revision).
2. Stephenson, H. Lance CCP, Editor, (2015). *Total Cost Management Framework: An Integrated Approach to Portfolio, Program and Project Management*, AACE International, 2nd Ed, Morgantown, WV.
3. AACE International, Recommended Practice 100R-19, *Contract Change Management as Applied in Engineering, Procurement, and Construction*, AACE International, Morgantown, WV (latest revision).
4. AACE International, Recommended Practice 25R-03, *Estimating Lost Labor Productivity in Construction Claims*, AACE International, Morgantown, WV (latest revision).
5. Long, Richard J. et al, (2014). *Cumulative Impact and Other Disruption Claims in Construction*, Virtualbookworm.com Publishing Inc., College Station, TX, USA.
6. Warhoe, Stephen P., (2012). *Applying Earned Value Management to Design-Bid-Build Projects to Assess Productivity Disruption: A System Dynamic Approach*, Dissertations.com Publishing, Boca Raton, FL, USA.
7. Callahan, Michael T, *Construction Change Order Claims*, 2nd ed., Construction Law Library, Wolters Kluwer, Aspen Publishers, 2005.
8. AACE Recommended Practice 120R-21, *Demonstrating Entitlement for Contract Change Orders or Claims – As Applied in Engineering, Procurement, and Construction*, AACE International, Morgantown, WV (latest revision).
9. Finke, Michael R., *Claims for Construction Productivity Losses*, 26 Pub. Contr. L.J. 311, page 312.
10. Nelson, Derek, *The Analysis and Valuation of Disruption*, 2011, adapted from Ibbs, Nguyen, and Lee, *Quantified Impacts of Project Change*, *Journal of Professional Issues in Engineering Education and Practice*, ASCE, January 2007.
11. *Change Orders Productivity Overtime – A Primer for the Construction Industry*, MCAA, 2012, p. 82, citing Centex Bateson Constr. Co., VABCA No. 4613, 5162, 5165, 99-1 BCA ¶ 30,153.

Legal References

1. Centex Bateson Constr. Co., VABCA Nos. 4613, 5162, 5165, 99-1 BCA ¶ 30,153 (1998), *aff'd*, Centex Bateson Constr. Co. v. West, 250 F.3rd 761 (Fed Cir. 2000).
2. McMillin Bros. Constructors, Inc., EBCA No. 328-10-84, 91-1 BCA ¶ 23,351.
3. Jackson Const. Co., Inc. v. United States, 62 Fed. Cl. 84, 103 (2004).
4. Appeal of Bechtel Nat., Inc., NASABCA No. 1186-7, 90-1 BCA ¶ 22,549 at 113,177 (1989).
5. Triple "A" South, 94-3. BCA ¶ 27,194, ASBCA No. 46,866 (1994) at 135,523.
6. Fruehauf Corp., PSBCA no. 477, 74-1 BCA ¶ 10,596 (1974).